1. Introduction to Control Systems

Control system = an interconnection of components forming a system configuration which gives a desired system response. In simple words: it **controls the output** of a process by manipulating the input so that output follows a desired value (reference).

Basic terms (write these in every answer)

- **System** – a combination of components acting together to perform a function.
- **Plant / Process** – the physical object to be controlled (motor, furnace, aircraft).
- **Input (reference) $r(t)$** – the desired value; **Output $c(t)$** – the actual controlled variable.
- **Controller** – generates the control signal from the error. **Actuator** – drives the plant.
- **Sensor / Feedback element** – measures output and sends it back. **Error $e(t) = r(t) - b(t)$.**
- **Disturbance** – unwanted signal that affects output (load change, wind, noise).

Examples

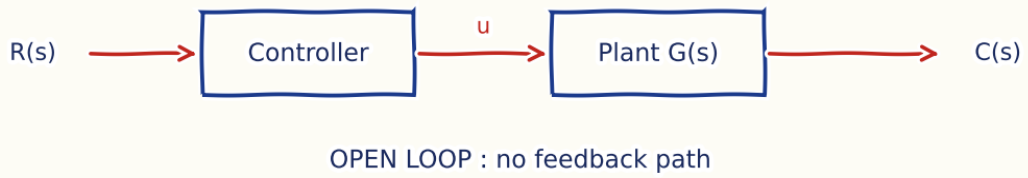
Room air-conditioner (thermostat), water-level control in a tank, speed control of a DC motor, automatic washing machine (open loop), traffic light (open loop), missile guidance, robot arm, cruise control in a car.

> ★ **Aim of a control system: (i) accuracy (ii) stability (iii) fast response (iv) rejection of disturbances (v) low sensitivity to parameter variations.**

2. Open Loop and Closed Loop Systems

Open loop system

*The control action is **independent of the output**. There is no feedback; output is neither measured nor compared with the input.*

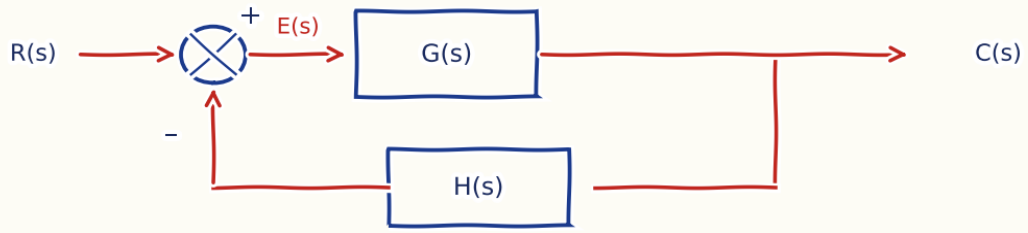


OPEN LOOP : no feedback path

***Example:** traffic signal (timer based), automatic washing machine, electric toaster, immersion rod with timer.*

Closed loop (feedback) system

*The control action depends on the output. Output is measured, fed back and compared with the reference. The difference (error) actuates the controller to reduce the error. Also called **feedback control system**.*



CLOSED LOOP : feedback H(s)

***Example:** room temperature control by thermostat, human being (eye–hand), speed governor, servo-mechanism, autopilot.*

Closed loop transfer function (negative feedback): $C(s)/R(s) = G(s) / [1 + G(s)H(s)]$

Positive feedback: $C(s)/R(s) = G(s) / [1 - G(s)H(s)]$

Differences — Open loop vs Closed loop (most asked 5M/10M)

S.No	Open Loop	Closed Loop
1	No feedback; output has no effect on control action	Feedback present; output affects control action
2	Simple, cheap, easy to design	Complex, costly, more components
3	Less accurate, accuracy depends on calibration	Highly accurate (error is corrected)
4	Stable (no stability problem)	May become unstable if gain is high
5	Sensitive to parameter variation & disturbance	Less sensitive to disturbance and parameter changes
6	Bandwidth is small	Bandwidth is large
7	Reliable for known, constant conditions	Used when conditions vary
8	Eg: traffic signal, toaster	Eg: AC thermostat, servo motor

3. Classification of Control Systems

Basis	Types	Remarks
Feedback	Open loop / Closed loop	With or without feedback path
Linearity	Linear / Non-linear	Linear obeys superposition: additivity + homogeneity
Time variation	LTI / LTV	LTI: parameters constant with time; LTV: vary with time (rocket mass)
Signal	Continuous-time / Discrete-time	Discrete uses sampled data, z-transform
Inputs–outputs	SISO / MIMO	Single / multiple input–output
Parameters	Lumped / Distributed	ODE vs PDE (transmission line)
Nature	Deterministic / Stochastic	Predictable vs random response
Purpose	Regulator / Servo / Process	Regulator keeps output constant; servo follows a varying reference

Superposition principle: *if* $r_1 \rightarrow c_1$ *and* $r_2 \rightarrow c_2$ *then* $(a\, r_1 + b\, r_2) \rightarrow (a\, c_1 + b\, c_2)$. *Systems with saturation, dead-zone, backlash, relay, friction are* ***non-linear***.

4. Feedback Characteristics and Effects of Feedback

Consider a negative feedback system with forward gain G and feedback gain H. Overall gain:

$$M = C/R = G / (1 + GH)$$

(a) Effect on overall gain

$|1 + GH| > 1$ *for negative feedback, hence overall gain is* ***reduced*** *by the factor* $(1 + GH)$. *For positive feedback gain increases but stability is lost.*

(b) Effect on sensitivity (parameter variation)

Sensitivity of M with respect to G = fractional change in M per fractional change in G:

$$S^{M}_{G} = (\partial M/M)/(\partial G/G) = 1 / (1 + GH)$$
$$S^{M}_{H} = - GH / (1 + GH)$$

*If $GH >> 1 : S^{M}_{G} \approx 0 \rightarrow$ the system is almost **insensitive** to variations in G (plant) but sensitive to H, so use a precise feedback element (sensor).*

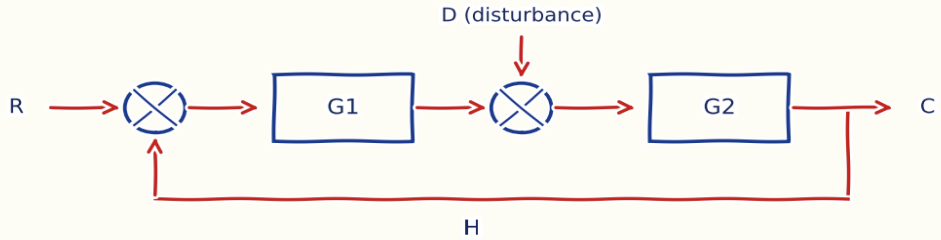
(c) Effect on stability

Feedback can make a stable open loop unstable (gain too high, phase lag at high frequency). The characteristic equation 1 + G(s)H(s) = 0 decides stability. Right choice of gain keeps system stable, and feedback can also stabilise an unstable plant.

(d) Effect on bandwidth

*Feedback **increases bandwidth** (gain × bandwidth ≈ constant), giving faster response (smaller rise time).*

(e) Effect on disturbance and noise



$$C = [G1G2 /(1+G1G2H)]\ R + [G2 /(1+G1G2H)]\ D$$

*With large G1H the effect of disturbance D on the output is reduced by (1 + G1G2H), whereas in open loop the disturbance appears fully at the output. Hence feedback gives **disturbance rejection** and reduces noise in the loop.*

(f) Other effects

- Reduces **steady-state error** (improves accuracy).
- Reduces non-linear distortion and effects of parameter drift.
- Improves transient response (can be tuned using gain), but may introduce oscillations.
- Increases complexity and cost; needs a sensor.

> ★ **Short answer: Feedback reduces gain, sensitivity, noise and error; it increases bandwidth, accuracy and disturbance rejection; but may reduce stability.**

5. Mathematical Models of Physical Systems

To analyse a system we first obtain its **mathematical model**, a set of equations that represents its dynamics. Methods: (i) differential equations (time domain) (ii) transfer function (s-domain) (iii) state-space model (Unit-V).

Differential equation model

Use physical laws: Newton's law (mechanical), Kirchhoff's laws (electrical), thermal and fluid laws. A general n^{th} order LTI system:

$$a_n \, d^n c/dt^n + \ldots + a_1 \, dc/dt + a_0 \, c = b_m \, d^m r/dt^m + \ldots + b_0 \, r$$

Transfer function (T.F.)

Definition: the ratio of the Laplace transform of output to the Laplace transform of input, with **all initial conditions zero**, for an LTI system.

$$G(s) = C(s)/R(s) = (b_m s^m + \ldots + b_0) \, / \, (a_n s^n + \ldots + a_0)$$

- Roots of numerator = **zeros**, roots of denominator = **poles**; denominator = characteristic polynomial; order n = highest power of s in denominator.
- T.F. is independent of input; applicable only to LTI systems; gives no information about the physical structure.

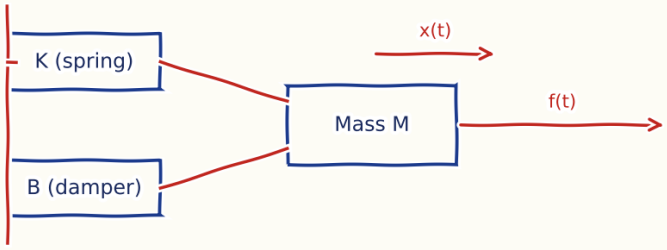
- T.F. = Laplace transform of the **impulse response**.
- Advantages: algebraic manipulation, easy stability and response analysis. Limitation: zero initial conditions, not for non-linear / time-varying / MIMO (conveniently).

Laplace pair	f(t)	F(s)
Unit step	$u(t)$	$1/s$
Unit ramp	t	$1/s^2$
Impulse	$\delta(t)$	1
Exponential	e^{-at}	$1/(s+a)$
Sine / Cosine	$\sin \omega t$ / $\cos \omega t$	$\omega/(s^2+\omega^2)$, $s/(s^2+\omega^2)$
Derivative	df/dt	$sF(s) - f(0)$
Integral	$\int f\, dt$	$F(s)/s$

6. Translational Mechanical Systems

Three basic elements (Newton's 2nd law: $\Sigma F = M\, a$):

Element	Force relation	Laplace form
Mass M	$f = M\, d^2x/dt^2$	$F(s) = M\, s^2\, X(s)$
Viscous friction (damper) B	$f = B\, dx/dt$	$F(s) = B\, s\, X(s)$
Spring K	$f = K\, x$	$F(s) = K\, X(s)$



Example 1: Mass–spring–damper

Applied force $f(t) = M\,x'' + B\,x' + K\,x$. Taking Laplace transform with zero initial conditions:

$$X(s)/F(s) = 1 / (M s^2 + B s + K)$$

Writing in standard form: $\omega_n = \sqrt{(K/M)}$, $\zeta = B / (2\sqrt{(KM)})$.

Example 2: Two masses (M1, M2) with springs K1, K2 and dampers B1, B2

Write one equation per mass (node), the relative motion terms use $(x1 - x2)$:

$$M1\,x1'' + B1\,x1' + K1\,x1 + B2\,(x1' - x2') + K2\,(x1 - x2) = f(t)$$

$$M2\,x2'' + B2\,(x2' - x1') + K2\,(x2 - x1) + B3\,x2' = 0$$

★ Procedure: (1) assume direction of x (2) draw free-body diagram for each mass (3) write $\Sigma F = Ma$ (4) Laplace transform (5) solve for X(s)/F(s) using Cramer's rule.

7. Rotational Mechanical Systems

Element	Torque relation	Laplace form
Moment of inertia J	$T = J\,d^2\theta/dt^2$	$T(s) = J\,s^2\,\theta(s)$
Rotational friction B	$T = B\,d\theta/dt$	$T(s) = B\,s\,\theta(s)$
Torsional spring K	$T = K\,\theta$	$T(s) = K\,\theta(s)$

Newton's law for rotation: $\Sigma\tau = J\,\alpha$. Example: a disc with inertia J, shaft stiffness K, bearing friction B and applied torque T(t):

$$T(t) = J\,\theta'' + B\,\theta' + K\,\theta \Rightarrow \theta(s)/T(s) = 1 / (J s^2 + B s + K)$$

For gear trains, reflect parameters to one shaft: $J_{eq} = J_1 + (N_1/N_2)^2 J_2$, $B_{eq} = B_1 + (N_1/N_2)^2 B_2$, where N is number of teeth $(\theta_2/\theta_1 = N_1/N_2)$.

8. Electrical–Mechanical Analogies

Mechanical (translational)	Force–Voltage analogy (series RLC)	Force–Current analogy (parallel RLC)
Force f	Voltage e	Current i
Mass M	Inductance L	Capacitance C
Friction B	Resistance R	Conductance $1/R$
Spring K	Elastance $1/C$	$1/L$
Displacement x	Charge q	Flux ψ
Velocity v	Current i	Voltage e

Rotational: torque T ↔ voltage (or current), J ↔ L (or C), B ↔ R (or $1/R$), K ↔ $1/C$ (or $1/L$), θ ↔ q. Analogy lets us study a mechanical system on an electrical circuit (and vice-versa). Series mechanical connection ↔ parallel electrical in F–I analogy.

9. Transfer Function of DC Servo Motor

A DC servo motor is a DC motor used in a feedback position/speed control system; it has low inertia, high torque/inertia ratio and linear torque–speed behaviour. Two types: ***armature controlled*** *and* ***field controlled****.*

(A) Armature controlled DC motor (field current constant)

Symbols: R_a, L_a armature resistance & inductance; e_a applied voltage; e_b back emf; K_t torque constant; K_b back-emf constant; J, B motor inertia, friction; θ shaft angle.

$$\textbf{Armature circuit: } e_a = i_a R_a + L_a \frac{di_a}{dt} + e_b$$

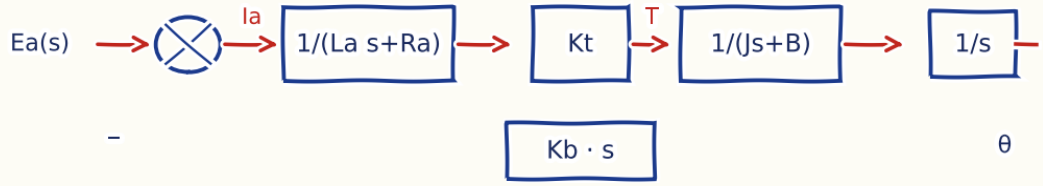
$$\textbf{Back emf: } e_b = K_b \frac{d\theta}{dt}$$

$$\textbf{Torque: } T = K_t i_a$$

$$\textbf{Mechanical: } T = J \frac{d^2\theta}{dt^2} + B \frac{d\theta}{dt}$$

Laplace transform (zero initial conditions):

$$E_a - K_b s\theta = (R_a + L_a s) I_a \; ; \; K_t I_a = (Js^2 + Bs)\theta$$



$$\theta(s)/E_a(s) = K_t / \{ s [(R_a + L_a s)(Js + B) + K_t K_b] \}$$

$$\text{If } L_a \approx 0 : \theta(s)/E_a(s) = K_m / [s (\tau_m s + 1)]$$

$$K_m = K_t / (R_a B + K_t K_b) , \tau_m = R_a J / (R_a B + K_t K_b)$$

*K_m = motor gain constant, τ_m = motor time constant. Note: back emf K_b acts like **internal feedback** that adds to damping (effective friction $B + K_t K_b / R_a$).*

(B) Field controlled DC motor (armature current constant)

Field circuit: $e_f = R_f i_f + L_f di_f/dt$. Torque $T = K_f i_f$ (flux $\propto i_f$, I_a constant), $T = J\theta'' + B\theta'$.

$$\theta(s)/E_f(s) = K_f / \{ s (L_f s + R_f)(Js + B) \} \text{ (third order)}$$

$$= K_m / \{ s (\tau_f s + 1)(\tau_m s + 1) \} , \tau_f = L_f/R_f , \tau_m = J/B$$

Point	Armature controlled	Field controlled
Order of T.F.	2nd (La neglected)	3rd
Constant quantity	Field current	Armature current
Efficiency / use	Better, widely used	Poor, small motors only
Inherent damping	Yes (back emf)	No

10. Transfer Function of AC Servo Motor

A two-phase induction motor with ***two stator windings*** *90° apart: a* ***reference winding*** *(fixed ac voltage) and a* ***control winding*** *(variable ac voltage* e_c*, 90° out of phase). Rotor has high resistance so the torque–speed curve is linear with negative slope (good damping, no run-away).*

Linearised torque equation around operating point:

$$T_m = K_1 e_c - K_2 \, d\theta/dt \; (K_1 = \text{control phase constant}, K_2 = \text{slope of } T\text{–}\omega \text{ curve})$$

$$T_m = J \, d^2\theta/dt^2 + B \, d\theta/dt$$

$$\square(s)/E_c(s) = K_1 / \{ s [Js + B + K_2] \} = K_m / [s (\tau_m s + 1)]$$

$$K_m = K_1 / (B + K_2) \, , \; \tau_m = J / (B + K_2)$$

- *Used where low power (< 1 hp) is enough: instruments, radar, small robots.*
- *Advantages: no commutator/brush, rugged, low maintenance.*
  Disadvantage: lower efficiency, non-linear characteristics, low torque.

> ★ ***AC servo vs DC servo: AC – brushless, low power, 50/400 Hz, lower efficiency; DC – brushed, high power, more efficient, commutator maintenance required.***

11. Synchro Transmitter and Receiver

A ***synchro*** *is an electromagnetic transducer (rotary transformer) that converts angular position of a shaft into an electrical signal; used as an* ***error detector*** *in servo systems. A synchro pair (transmitter + control transformer/receiver) measures angular difference between two shafts.*

Synchro transmitter (Tx)

Construction: a **rotor** (dumb-bell, single winding on slip rings, excited by a.c. voltage $V_r \sin \omega t$) and a **stator** with three identical star-connected windings S1, S2, S3 spaced 120° apart. The rotor flux induces emfs in the stator windings, magnitude depending on rotor angle θ.

$$e_{S1n} = K V_r \sin \omega t \cos \theta$$
$$e_{S2n} = K V_r \sin \omega t \cos(\theta - 120°)$$
$$e_{S3n} = K V_r \sin \omega t \cos(\theta + 120°)$$

Line voltages: $e_{S1S2} = \sqrt{3}\, K V_r \sin \omega t \sin(\theta + 240°)$

$e_{S2S3} = \sqrt{3}\, K V_r \sin \omega t \sin \theta$, $e_{S3S1} = \sqrt{3}\, K V_r \sin \omega t \sin(\theta + 120°)$

Maximum line voltage ($\sqrt{3}\, K V_r$) is obtained when rotor is at 90° with respect to the stator coil axis; at θ = 0°, 120°, 240° one of the line voltages is zero. These 3 line voltages give a **unique** set for every angle → position information.

Synchro receiver / control transformer (CT) as error detector

The stator of the control transformer is connected to the Tx stator. Rotor of CT is not excited; the voltage across rotor terminals is the output:

$$e_0(t) = K' \sin \omega t \cdot \sin(\theta_t - \theta_r) \approx K' (\theta_t - \theta_r) \sin \omega t \text{ for small error}$$

So output is a suppressed-carrier amplitude-modulated signal proportional to the **error** $\theta_e = \theta_t - \theta_r$. It is demodulated and amplified to drive a servo motor. Transfer function of the error detector:

$$E_0(s)/\square_e(s) = K_s \text{ (sensitivity of synchro, volt/rad} \approx \text{volts per degree)}$$

Torque synchro (receiver) type: the rotor of receiver is also excited; Tx and Rx stators joined; if rotor positions differ a torque acts to align them. Equilibrium (zero torque) when both rotor angles are equal. Receiver rotor

has a damper to avoid oscillations.

- *Advantages: robust, no contact (except slip rings), accurate (± 0.1°), operates over long distances.*
- *Applications: remote position indication, radar antenna, gun-turret, robot arms, machine tools.*

12. Quick Revision / Important Questions

- *Define control system; differentiate open and closed loop with examples. (5M)*
- *Explain the effects of feedback on gain, sensitivity, stability, bandwidth, noise. (10M)*
- *Derive T.F. of armature controlled DC servo motor with block diagram. (10M)*
- *Derive the T.F. of field controlled DC motor. Compare with armature control. (10M)*
- *Write the differential equations for the given mechanical system and draw force–voltage and force–current analogous circuits. (10M)*
- *Explain the construction and working of synchro transmitter and receiver; derive the line voltages. (10M)*
- *Derive the T.F. of AC servo motor from its torque–speed characteristic. (5M)*
- *Define transfer function; list its properties/limitations. (2M/5M)*

$$C/R = G/(1+GH) \mid S^{M}_{G} = 1/(1+GH) \mid X/F = 1/(Ms^2+Bs+K)$$

$$\text{Armature motor: } K_t/\{s[(R_a+L_a s)(Js+B)+K_t K_b]\} \mid \text{AC servo: } K_m/[s(\tau_m s+1)]$$

★ ***Exam tip: always draw the block diagram / free-body diagram first, state assumptions (zero initial conditions, L_a neglected) and box the final transfer function.***

1. Block Diagram Representation

A **block diagram** is a pictorial representation of the functions performed by each component and of the flow of signals. It shows the relationship between input and output of every element through its transfer function.

Elements of a block diagram

- **Block** – rectangle with the T.F. G(s) inside; output = G(s) × input. Arrow = direction of signal flow.
- **Summing point** – circle with ⊠ ; signs (+/−) show addition or subtraction of signals.
- **Take-off (branch) point** – dot from which the same signal goes to more than one place; value does not change.
- **Forward path** – from input to output; **feedback path** – from output back to the summing point.

Open loop T.F. = $G(s)H(s)$, Feed-forward T.F. = $C(s)/E(s) = G(s)$

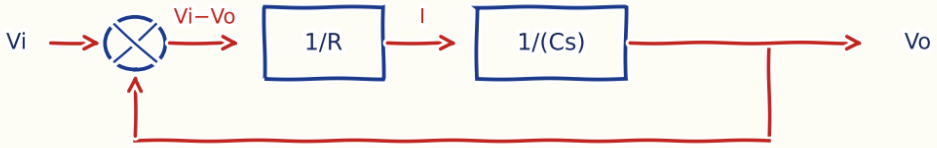
Closed loop T.F. : $C(s)/R(s) = G(s)/[1 + G(s)H(s)]$; $E(s)/R(s) = 1/[1 + G(s)H(s)]$

Procedure to draw the block diagram of an electrical system

- Write the differential equations of the network (KVL / KCL).
- Take Laplace transform with zero initial conditions.
- Rearrange each equation as (output of block) = (T.F.) × (input), choosing one variable per block.
- Connect blocks using summing points and take-off points; check that every variable appears once as an output.

Example: Series RC circuit (input V_i, output V_o across C)

$$I(s) = [V_i(s) - V_o(s)] / R \; ; \; V_o(s) = I(s) / (Cs)$$



Using the feedback formula (G = 1/(RCs), H = 1):

$$V_o(s)/V_i(s) = [1/(RCs)] / [1 + 1/(RCs)] = 1 / (RCs + 1)$$

Example: Series R–L–C circuit (output across C)

Loop equations: $V_i - V_o = (R + Ls)\, I$ and $V_o = I/(Cs)$. So we use a block $1/(R+Ls)$ (error voltage → current), a block $1/(Cs)$ (current → V_o) and unity feedback:

$$V_o/V_i = 1 / (LC\, s^2 + RC\, s + 1)$$

2. Block Diagram Algebra (Reduction Rules)

Rule	Original	Equivalent
1. Cascade blocks	$G1 \rightarrow G2$	$G1 \cdot G2$
2. Parallel blocks	$G1 \pm G2$ (summed)	$G1 \pm G2$
3. Feedback loop (neg.)	G with feedback H	$G / (1 + GH)$
4. Shift summing point beyond a block	$R \rightarrow (+) \rightarrow G$, extra input X before G	X must be multiplied by G after block $(X \cdot G)$
5. Shift summing point ahead of a block	Extra input X after G	X must be divided by G (X/G) before block
6. Shift take-off point beyond a block	Take-off before G	Take-off after G, multiply by 1/G in tapped branch
7. Shift take-off point ahead of a block	Take-off after G	Take-off before G, multiply tapped branch by G
8. Interchange summing points	$a \pm b \pm c$	Order can be changed freely
9. Eliminate feedback H	Feedback path H(s)	Unity feedback with $G/(1+G(H-1))$ approach: put 1/H before the loop and H after
10. Move take-off across summing point	–	Not allowed directly; add a summing point on the branch

★ Golden rule of reduction: the product of transfer functions in the forward path must remain the same, and the product around any loop must remain the same.

Steps for reducing a complicated block diagram

- Combine all **cascade** blocks. Combine all **parallel** blocks.
- Eliminate all **inner feedback loops**.
- Shift summing points to the left and take-off points to the right of the loop.
- Repeat steps 1–3 until a single block remains.

Worked example

Forward path G1 → G2 → G3, inner feedback H1 around G2, outer feedback H2 from output to input summing point.

Step 1: inner loop = G2/(1+G2H1). Step 2: cascade = G1·[G2/(1+G2H1)]·G3. Step 3: outer loop:

C/R = G1G2G3 / (1 + G2H1 + G1G2G3H2)

3. Signal Flow Graph (SFG)

*An SFG is a graphical representation of a set of linear algebraic equations. It consists of **nodes** (variables) joined by directed **branches** (gains = transfer functions). It is simpler than block diagram reduction because Mason's gain formula gives the T.F. directly.*

Terminology

Term	Meaning
Node	Represents a system variable (sum of incoming signals)
Branch	Directed line segment between nodes with gain (transmittance)
Input (source) node	Only outgoing branches
Output (sink) node	Only incoming branches
Mixed node	Both incoming and outgoing branches
Path	Traversal of branches in the direction of arrows without repeating a node
Forward path	Path from input node to output node, no node crossed twice
Loop	Closed path starting and ending at the same node, no node crossed twice
Self loop	Loop with one branch and one node
Non-touching loops	Loops having no common node
Path gain / Loop gain	Product of branch gains along the path / loop

Properties of SFG

- Applicable to linear systems only; node value = sum of all incoming signals.
- Signal flows only in the direction of the arrow; a branch multiplies signal by its gain.
- Equations must be written as effect = Σ (gain $\times$ cause).

Converting a block diagram to SFG

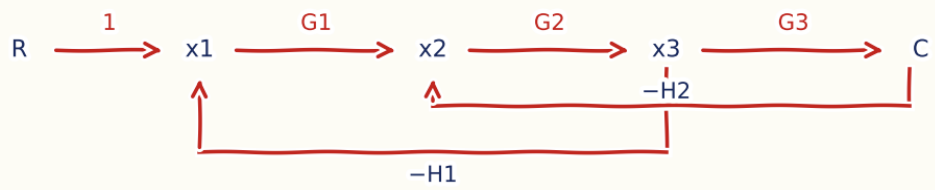
- Assign a node to the input, the output, every summing point and every take-off point.
- Draw a branch for each block with gain = T.F.; branch with gain -1 or $-H$ for negative feedback.
- Summing point node = sum of incoming branches (with their signs).

4. Mason's Gain Formula

$$T = C(s)/R(s) = (\Sigma_k P_k \Delta_k) / \Delta$$

- P_k = gain of the kth forward path.
- $\Delta = 1 -$ (sum of all individual loop gains) + (sum of gain products of all possible pairs of non-touching loops) $-$ (sum of products of all possible triplets of non-touching loops) + ...
- Δ_k = value of Δ after removing all loops touching the kth forward path (put these loop gains to zero).

Example 1



Forward path: P_1 = G1G2G3 (only one). Loops: L_1 = $-$G1G2H1 (x1→x2→x3→x1), L_2 = $-$G2G3H2 (x2→x3→C→x2). Both loops share node x2 → **touching**, so no

non-touching products.

$$\Delta = 1 - (L1 + L2) = 1 + G1G2H1 + G2G3H2 \; ; \Delta_1 = 1$$

$$T = G1G2G3 \; / \; (1 + G1G2H1 + G2G3H2)$$

Example 2 (with non-touching loops)

Let a system have two forward paths $P_1 = G1G2G3$ and $P_2 = G4$, loops $L_1 = -G2H1$, $L_2 = -G3H2$ and $L_3 = -G1G2G3H3 - \ldots$; suppose L_1 and L_2 do not touch each other.

$$\Delta = 1 - (L1 + L2 + L3) + (L1 \cdot L2)$$

Path P1 touches all loops → $\Delta_1 = 1$; Path P2 touches none of L1, L2 → $\Delta_2 = 1 - L3$ (if L3 touches P2 then $\Delta_2 = 1 -$ (loops not touching P2))

$$T = (P1 \cdot \Delta 1 + P2 \cdot \Delta 2) / \Delta$$

★ **Common mistakes: forgetting non-touching loop pair products, taking wrong sign of loop gains (negative feedback gives negative loop gain), and not recalculating Δk for each path.**

5. Compensation Techniques

*A **compensator** is an additional network/controller added to the system so that the performance specifications (steady-state error, peak overshoot, settling time, gain/phase margin) are satisfied when gain adjustment alone is not enough.*

Type	Position	Used when
Series (cascade) compensation	In forward path, before the plant	Most common; easy design
Feedback (parallel) compensation	In a feedback path around part of the plant	Reduces effect of parameter variations
Feed-forward compensation	Parallel path from input	Improves disturbance rejection/speed

Lead compensator

Passive RC lead network (R_1 in parallel with C, in series with R_2):

$$G_c(s) = (1/\alpha)\,(1 + \alpha Ts)/(1 + Ts)\,,\ \alpha = (R1 + R2)/R2 > 1\,,\ T = R1R2C/(R1 + R2)$$

$$\text{Maximum phase lead: } \varphi_m = \sin^{-1}[(\alpha - 1)/(\alpha + 1)] \text{ at } \omega_m = 1/(T\sqrt{\alpha})$$

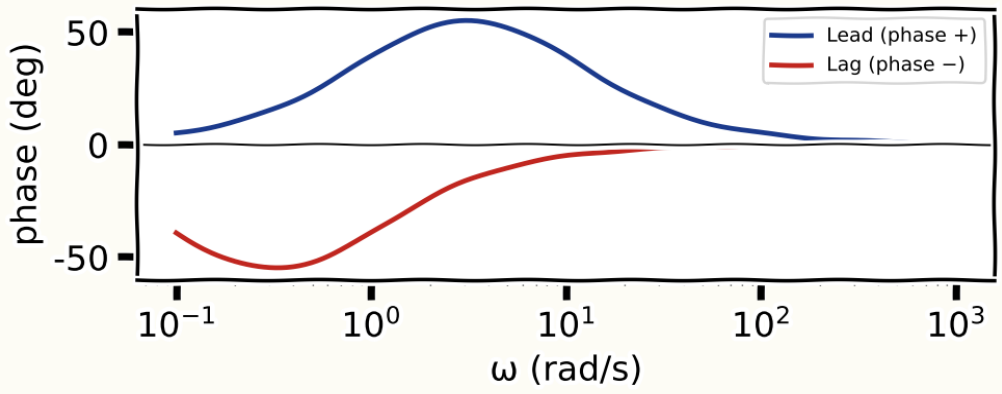
- *Zero lies closer to origin than pole (zero at $1/\alpha T$, pole at $1/T$). Acts like a high-pass filter / derivative action.*
- ***Effects:*** *adds positive phase → increases phase margin, damping and stability; increases bandwidth → faster response (lower rise & settling time); amplifies high-frequency noise; does not improve steady-state accuracy much.*
- *Used when the system needs better* ***transient response****.*

Lag compensator

$$G_c(s) = (1 + Ts)/(1 + \beta Ts)\,,\ \beta = (R1 + R2)/R2 > 1\,,\ T = R2C$$

$$\text{Maximum phase lag: } \varphi_m = \sin^{-1}[(1 - \beta)/(1 + \beta)] \text{ at } \omega_m = 1/(T\sqrt{\beta})$$

- *Pole is closer to origin than the zero. Acts like a low-pass filter / integral action.*
- ***Effects:*** *reduces high-frequency gain (gain crossover moves to lower ω); improves steady-state accuracy (increases error constants K_p, K_v by β factor); decreases bandwidth → slower response; reduces noise; stability margin gets better by lowering ω_{gc} but phase lag is added.*
- *Used when the system needs better* ***steady-state accuracy*** *with acceptable transient behaviour.*



Lag–Lead compensator

$$G_c(s) = [\,(1 + T1\,s)(1 + T2\,s)\,] / [\,(1 + (T1/\gamma)\,s)(1 + \gamma\,T2\,s)\,],\ \gamma > 1$$

(lead part: T1 s terms; lag part: T2 s terms)

Combines the advantages of both: lag part improves steady-state accuracy, lead part improves transient response and stability. Used when both are required.

Parameter	Lead	Lag	Lag–Lead
Pole/Zero	zero nearer origin	pole nearer origin	both
Phase	+ve	−ve (small at design ω)	−ve at low ω, +ve at high ω
Bandwidth	increases	decreases	increases
Rise/settling time	decreases	increases	decreases
Steady-state error	small change	reduces	reduces
Noise	amplified	attenuated	–
Like	PD controller	PI controller	PID controller

6. Design of Compensators (Bode Plot Method)

Lead compensator design steps

- Using the given static error constant (K_v or K_p) determine gain K; plot Bode of KG(s).

- Find the existing phase margin (PM) and compute the extra phase needed: $\varphi_m = PM_{desired} - PM_{existing} + (5° \text{ to } 12° \text{ safety})$.
- Calculate $\alpha = (1 + \sin \varphi_m)/(1 - \sin \varphi_m)$.
- Find frequency where $|KG(j\omega)| = -10 \log \alpha$ dB; take it as new gain crossover ω_m.
- $T = 1/(\omega_m \sqrt{\alpha})$; compensator = $(1 + \alpha Ts)/(1 + Ts)$; restore the gain loss ($K_c = \alpha$, if the network has $1/\alpha$ attenuation).
- Check PM, GM of the compensated system; repeat if specs are not met.

Lag compensator design steps

- Set K from steady-state requirement; draw Bode of KG(s).
- Find the frequency ω_c at which phase = $-180° + PM_{desired} + (5° \text{ to } 12°)$ — new gain crossover.
- Measure gain of $KG(j\omega_c)$ in dB, set $20 \log \beta$ = that value → β.
- Place the zero one decade below ω_c: $1/T = \omega_c/10$; pole at $1/(\beta T)$.
- $Gc(s) = (1 + Ts)/(1 + \beta Ts)$; verify the margins.

Lag–lead design (outline)

Choose γ from the required K_v and the new gain crossover frequency; design lead part for phase margin; then the lag part (T_2) with zero at $\omega_c/10$; combine.

> ★ **Exam trick: α for lead is > 1 and β for lag is > 1; always check that the phase of the lag network at ω_c is only a few degrees ($-5°$ to $-12°$).**

7. PID Controllers

The **PID controller** produces a control signal from the error e(t) by Proportional, Integral and Derivative actions:

$$u(t) = K_p\, e(t) + K_i \int e\, dt + K_d\, de/dt$$

$$G_c(s) = K_p + K_i/s + K_d\, s = K_p[\,1 + 1/(T_i\, s) + T_d\, s\,],\ K_i = K_p/T_i,\ K_d = K_p T_d$$

Action	T.F.	Effect	Drawback
P	K_p	Reduces rise time & steady-state error (not zero)	Offset remains, larger K_p → oscillation
I	K_i/s	Eliminates steady-state error (raises type by 1)	Slower, increases overshoot, may destabilise
D	$K_d s$	Adds damping, reduces overshoot, anticipates error	Amplifies noise, no effect on steady-state error
PI	$K_p + K_i/s$	Zero steady-state error to step; acts like lag	Slower transient
PD	$K_p + K_d s$	Faster transient; acts like lead	High-frequency noise

Effect of increasing each gain (step response)

Gain ↑	Rise time	Overshoot	Settling time	Steady-state error
K_p	Decrease	Increase	Small change	Decrease
K_i	Decrease	Increase	Increase	Eliminate
K_d	Small change	Decrease	Decrease	No change

Ziegler–Nichols tuning (ultimate-gain method)

Set I and D to zero, increase K_p until sustained oscillation: ultimate gain K_u and period P_u.

Controller	K_p	T_i	T_d
P	$0.5\,K_u$	∞	0
PI	$0.45\,K_u$	$P_u/1.2$	0
PID	$0.6\,K_u$	$P_u/2$	$P_u/8$

8. More Solved Problems (exam type)

Problem 1 – Mason's formula with a feed-forward path

Nodes: R → E (gain 1), E → X (G1), X → C (G2), C → E (−H), and a direct branch R → C (G3).

Forward paths: P1 = G1G2 , P2 = G3

Loop: L1 = −G1G2H ⇒ Δ = 1 + G1G2H

P2 shares node C with L1 → Δ1 = Δ2 = 1

T = C/R = (G1G2 + G3) / (1 + G1G2H)

Check by algebra: *C = G2G1E + G3R and E = R − HC ⇒ C(1 + G1G2H) = (G1G2 + G3) R ☒.*

Problem 2 – Two non-touching loops

Chain R →(G1)→ x1 →(G2)→ x2 →(G3)→ x3 →(G4)→ x4 →(G5)→ C. Feedbacks: −H1 from x2 to x1, −H2 from x4 to x3, −H3 from x4 to x2.

L1 = −G2H1 , L2 = −G4H2 , L3 = −G3G4H3

L1 and L2 do not touch (no common node) → product L1L2 = G2G4H1H2

Δ = 1 + G2H1 + G4H2 + G3G4H3 + G2G4H1H2

P1 = G1G2G3G4G5 touches all loops → Δ1 = 1

T = G1G2G3G4G5 / (1 + G2H1 + G4H2 + G3G4H3 + G2G4H1H2)

Problem 3 – Lead compensator design (numerical)

Plant G(s) = 4/[s(s+2)]. Specifications: $K_v = 20\ s^{-1}$, PM ≥ 50°.

- *K_v = lim sKG = 2K = 20 ⇒ K = 10, so KG(s) = 40/[s(s+2)].*
- *Gain crossover: $40/(\omega\sqrt{(\omega^2+4)}) = 1 \Rightarrow \omega_{gc} \approx 6.17$ rad/s; PM = 90° − $\tan^{-1}(6.17/2) \approx 18°$.*
- *Required lead: $\varphi_m = 50° - 18° + 5° = 37°$.*
- *$\alpha = (1 + \sin 37°)/(1 - \sin 37°) \approx 4.0$; extra gain at ω_m = 10 log α ≈ 6 dB.*

- Find ω where $|KG| = -6$ dB $\Rightarrow \omega_m \approx 8.84$ rad/s. $T = 1/(\omega_m\sqrt{\alpha}) \approx 0.0564$, $\alpha T \approx 0.227$.

$G_c(s) = (1 + 0.227s)/(1 + 0.0564s)$ (with gain K = 10 retained)
New PM $\approx 90° - \tan^{-1}(8.84/2) + 37° \approx 50°$ ☒ ; K_v unchanged = 20

Lag–lead RC network

$E_o/E_i = (1 + R1C1\,s)(1 + R2C2\,s) / [\,(1 + R1C1\,s)(1 + R2C2\,s) + R1C2\,s\,]$

Poles and zeros are real; the lag part acts at low frequencies and lead part at higher frequencies; at $\omega = 1/\sqrt{(R1C1R2C2)}$ the network phase is zero.

Block diagram vs Signal flow graph

Point	Block diagram	SFG
Representation	Blocks, summing & take-off points	Nodes and branches
Reduction	Step by step algebra (slow for big systems)	Mason's gain formula (direct)
Applicable to	Linear and non-linear (with blocks)	Linear systems only
Best for	Physical understanding of system	Quick overall T.F.

★ Remember: for the final answer always write the closed-loop T.F. in the form N(s)/D(s) and mention the characteristic equation D(s) = 0.

Important Questions

- State and explain the block diagram reduction rules with an example. (10M)
- Find C/R using Mason's gain formula for the given signal flow graph. (10M)
- Explain lead, lag and lag–lead compensators with circuits and T.F. (10M)
- Design a lead/lag compensator for the given specifications using Bode plot. (10M)
- Explain PID controller; effects of P, I, D actions; Z–N tuning. (5M/10M)

1. Standard Test Signals

The actual input to a system is often unknown, so its performance is compared using standard test signals whose Laplace transforms are simple.

Signal	r(t)	R(s)	Used to test
Step	$A\,u(t)$	A/s	Sudden change (position); transient specs
Ramp	$A\,t$	A/s^2	Constant-velocity inputs (tracking)
Parabolic	$A\,t^2/2$	A/s^3	Constant acceleration
Impulse	$A\,\delta(t)$	A	Sudden shock; gives the T.F. (impulse response)
Sinusoidal	$A \sin \omega t$	$A\omega/(s^2+\omega^2)$	Frequency response

Time response $c(t)$ = **transient** response (dies out as $t \to \infty$) + **steady-state** response (remains as $t \to \infty$).

2. Time Response of First Order Systems

$C(s)/R(s) = 1 / (T s + 1)$ (T = time constant, pole at $s = -1/T$)

Input	C(s)	c(t)	Remarks
Unit step	$1/[s(Ts+1)]$	$1 - e^{-t/T}$	$c(T) = 63.2\ \%$ of final value; no overshoot
Unit ramp	$1/[s^2(Ts+1)]$	$t - T + T\,e^{-t/T}$	$e_{ss} = T$ (lags the input)
Unit impulse	$1/(Ts+1)$	$(1/T)\,e^{-t/T}$	Decays to zero; derivative of step response

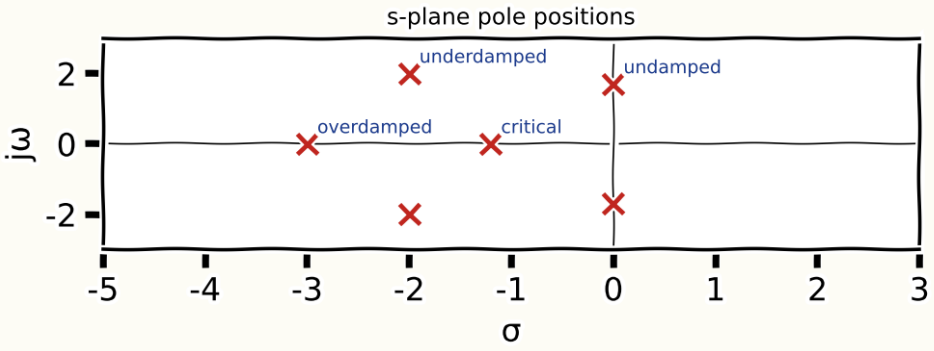
- **Time constant T:** time taken to reach 63.2 % of the final value for a step input.
- Settling time: $t_s = 4T$ (2 % band), $3T$ (5 % band); rise time t_r (10 %–90 %) = $2.2T$; delay time $t_d = 0.69T$.
- Smaller $T \rightarrow$ pole farther left $\rightarrow$ faster response. Response to derivative of input = derivative of response (LTI property).

3. Characteristic Equation of Feedback Control Systems

For the closed loop system $C/R = G/(1 + GH)$ the denominator set to zero is the **characteristic equation**:

$$\mathbf{1 + G(s)H(s) = 0 \Rightarrow roots = closed\ loop\ poles}$$

- The poles decide stability and nature of the transient response; zeros affect only the magnitude/shape.
- Poles in LHP $\rightarrow$ stable decaying terms; on imaginary axis $\rightarrow$ sustained oscillation; in RHP $\rightarrow$ growing.



4. Transient Response of Second Order Systems

$$C(s)/R(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$$

ω_n = natural frequency, ζ = damping ratio, $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$ = damped frequency

Poles: $s = -\zeta\omega_n \pm j\omega_n\sqrt{(1-\zeta^2)}$

For unity feedback with $G(s) = \omega_n^2/[s(s + 2\zeta\omega_n)]$ the closed loop gives the above T.F.

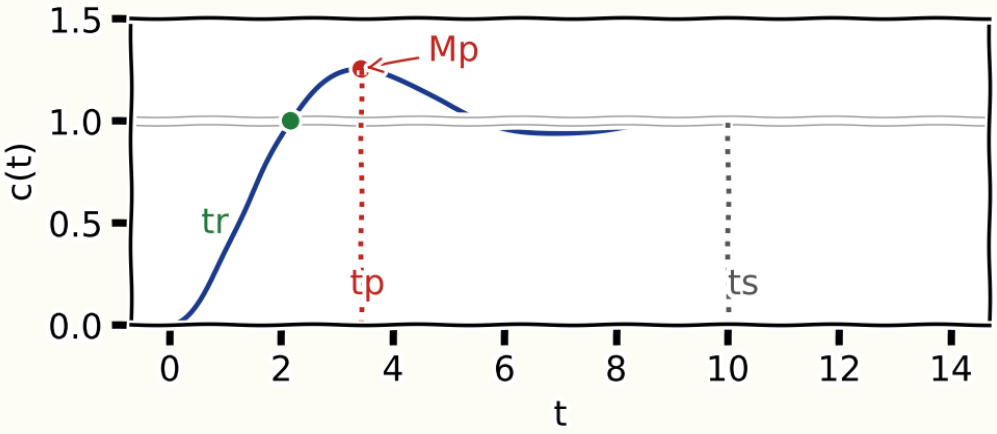
ζ	Type	Poles	Step response
0	Undamped	$\pm j\omega_n$ (imag. axis)	Sustained oscillations (constant amplitude)
$0 < \zeta < 1$	Underdamped	complex conjugate, LHP	Damped oscillations, overshoot
1	Critically damped	real, equal ($-\omega_n$)	Fastest non-oscillatory rise
> 1	Overdamped	real, unequal	Sluggish, no overshoot

Step response – underdamped ($0 < \zeta < 1$)

$$c(t) = 1 - [\, e^{-\zeta\omega_n t} / \sqrt{(1-\zeta^2)}\,] \sin(\omega_d t + \varphi)\,,\ \varphi = \cos^{-1}\zeta = \tan^{-1}[\sqrt{(1-\zeta^2)}/\zeta]$$

Critically damped: $c(t) = 1 - (1 + \omega_n t)\, e^{-\omega_n t}$. Overdamped: $c(t) = 1 + [1/(2\sqrt{(\zeta^2-1)}(\zeta+\sqrt{(\zeta^2-1)}))]\, e^{-(\zeta-\sqrt{(\zeta^2-1)})\omega_n t} - [1/(2\sqrt{(\zeta^2-1)}(\zeta-\sqrt{(\zeta^2-1)}))]\, e^{-(\zeta+\sqrt{(\zeta^2-1)})\omega_n t}$.

5. Time Domain Specifications



Specification	Definition	Formula (underdamped)
Delay time t_d	Time to reach 50 % of final value	$\approx (1 + 0.7\zeta)/\omega_n$
Rise time t_r	0→100 % (underdamped)	$t_r = (\pi - \varphi)/\omega_d$
Peak time t_p	Time of first peak	$t_p = \pi/\omega_d$
Peak overshoot M_p	$(c_{max} - c_{ss})/c_{ss} \times 100$	$M_p = e^{-\pi\zeta/\sqrt{(1-\zeta^2)}} \times 100$ %
Settling time t_s	Time to stay within 2 % (5 %) band	$t_s = 4/(\zeta\omega_n)$ $(3/(\zeta\omega_n))$

$$\zeta \text{ from } M_p:\ \zeta = -\ln(M_p/100) / \sqrt{(\pi^2 + \ln^2(M_p/100))}$$

Solved example

Unity feedback, $G(s) = 25/[s(s+5)]$. Closed loop: $\omega_n^2 = 25 \rightarrow \omega_n = 5$, $2\zeta\omega_n = 5 \rightarrow \zeta = 0.5$, $\omega_d = 5\sqrt{0.75} = 4.33$ rad/s, $\varphi = \cos^{-1}0.5 = 1.047$ rad.

$$t_r = (\pi - 1.047)/4.33 = 0.484\ s\ ;\ t_p = \pi/4.33 = 0.725\ s$$
$$M_p = e^{-\pi(0.5)/0.866} = 16.3\ \%\ ;\ t_s\ (2\ \%) = 4/(0.5\times5) = 1.6\ s$$

★ Higher $\zeta \rightarrow$ lower overshoot; higher ω_n (at same ζ) $\rightarrow$ faster response (smaller t_r, t_p, t_s). Desirable $\zeta = 0.4 - 0.707$.

6. Steady State Response and Steady State Errors

Steady-state error is the difference between input and output as $t \rightarrow \infty$. For unity feedback, $E(s) = R(s)/[1 + G(s)]$, by final value theorem:

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to0} s\,R(s) / [\,1 + G(s)H(s)\,]$$

System type and error constants

Write $G(s)H(s) = K(1+T_a s)\ldots / [\,s^N(1+T_1 s)\ldots\,]$; N = number of poles at origin = **type** of the system.

$$\text{Position error constant } K_p = \lim_{s\to0} G(s)H(s)$$
$$\text{Velocity error constant } K_v = \lim_{s\to0} s\,G(s)H(s)$$
$$\text{Acceleration error constant } K_a = \lim_{s\to0} s^2\,G(s)H(s)$$

Type	Step $e_{ss}=1/(1+K_p)$	Ramp $e_{ss}=1/K_v$	Parabolic $e_{ss}=1/K_a$
0	$1/(1+K)$	∞	∞
1	0	$1/K$	∞
2	0	0	$1/K$
3	0	0	0

Solved: $G(s)H(s) = 10(s+2)/[s^2(s+1)(s+5)]$ → *Type 2,* $K_p = \infty$, $K_v = \infty$, $K_a = 10 \cdot 2/(1 \cdot 5) = 4$ → e_{ss} *(step) = 0, (ramp) = 0, (parabolic with unit magnitude) =* $1/4 = 0.25$.

Effects of P-D and P-I controllers on the time response

- ***Proportional-Derivative (PD):*** $G_c = K_p(1 + T_d s)$ *adds a zero →* *increases damping* ($\zeta_{eff} = \zeta + \omega_n T_d/2$ *for the standard* 2nd*-order loop), reduces* M_p, t_r, t_s*; steady-state error for ramp unchanged (type unchanged); amplifies noise.*
- ***Proportional-Integral (PI):*** $G_c = K_p(1 + 1/(T_i s))$ *adds a pole at origin →* ***type increases by 1****, steady-state error reduces (step error becomes zero for type 0); but system becomes less stable and more oscillatory (*M_p *increases), slower response.*

7. Stability Analysis – Concept of Stability

A linear time-invariant system is ***stable*** *if for a bounded input the output is bounded (BIBO) and for zero input the output returns to zero from any initial condition.*

Condition	Location of closed-loop poles	Impulse response
Absolutely stable	All in left half s-plane	→ 0
Marginally (critically) stable	Simple poles on jω axis, none in RHP	sustained oscillation / constant
Unstable	Any pole in RHP or repeated poles on jω axis	grows without bound
Conditionally stable	Stable only for a limited range of gain K	–

- **Necessary condition:** all coefficients of the characteristic polynomial must be real, positive and non-zero (none missing). Not sufficient for order ≥ 3.
- **Relative stability:** how far the poles are from the jω axis (measured by shifting $s \to s - \sigma$, or gain/phase margins).

8. Routh–Hurwitz Stability Criterion

For $q(s) = a_0 s^n + a_1 s^{n-1} + \ldots + a_n = 0$ form the Routh array. **The system is stable iff all elements of the first column have the same sign**; the number of sign changes = number of roots in the RHP.

s^n	a_0	a_2	a_4
s^{n-1}	a_1	a_3	a_5
s^{n-2}	$b_1 = (a_1a_2 - a_0a_3)/a_1$	$b_2 = (a_1a_4 - a_0a_5)/a_1$	…
s^{n-3}	$c_1 = (b_1a_3 - a_1b_2)/b_1$	…	…
…	…	…	…
s^0	a_n		

Example 1: $s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$

s^4	1	3	5
s^3	2	4	0
s^2	$(6-4)/2 = 1$	$(10-0)/2 = 5$	0
s^1	$(4-10)/1 = -6$	0	0
s^0	5		

Signs: + + + − + → 2 sign changes → ***2 roots in RHP, unstable.***

Example 2: range of K for $s^3 + 6s^2 + 8s + K = 0$

Array: s^3: 1, 8 | s^2: 6, K | s^1: (48 − K)/6 | s^0: K. For stability K > 0 and 48 − K > 0:

0 < K < 48 (marginal at K = 48: $A(s) = 6s^2 + 48 = 0 \Rightarrow s = \pm j2.83$, so $\omega_{osc} = 2.83$ rad/s)

Special cases

- ***Case 1 – a zero appears in the first column*** *(other row elements non-zero): replace 0 by a small +ε, complete the array, then let ε → 0 and count sign changes. (Or substitute s = 1/z.)*

- **Case 2 – an entire row becomes zero**: *form the auxiliary equation A(s) from the row above, differentiate dA/ds, put its coefficients in the zero row, continue. Roots of A(s) are symmetrical about the origin ($\pm\sigma$ or $\pm j\omega$ pairs); if no sign change below, they are on $j\omega$ axis → marginally stable.*

Conditional stability

Some systems are stable only for a certain range of gain (e.g., $0 < K < 48$ above) or become stable only when K exceeds a minimum (e.g., higher-order systems stable for $K_1 < K < K_2$). Such a system is ***conditionally stable****; reducing gain below K_1 (for example due to saturation) can make it unstable.*

> ★ ***Limitations of Routh criterion: (1) applicable only to polynomials with real, constant coefficients (no transport delay e^{-sT} unless approximated); (2) gives no exact root locations; (3) does not indicate relative stability or how to stabilise the system; (4) not directly for non-polynomial/ irrational equations.***

9. Root Locus Concept

The ***root locus*** *is the path traced by closed-loop poles in the s-plane as a parameter (usually gain K, $0 \rightarrow \infty$) varies.*

Characteristic equation $1 + K\,G(s)H(s) = 0$ gives conditions:

Angle condition: $\angle G(s)H(s) = \pm(2q+1)180^\circ$, $q = 0, 1, 2, \ldots$

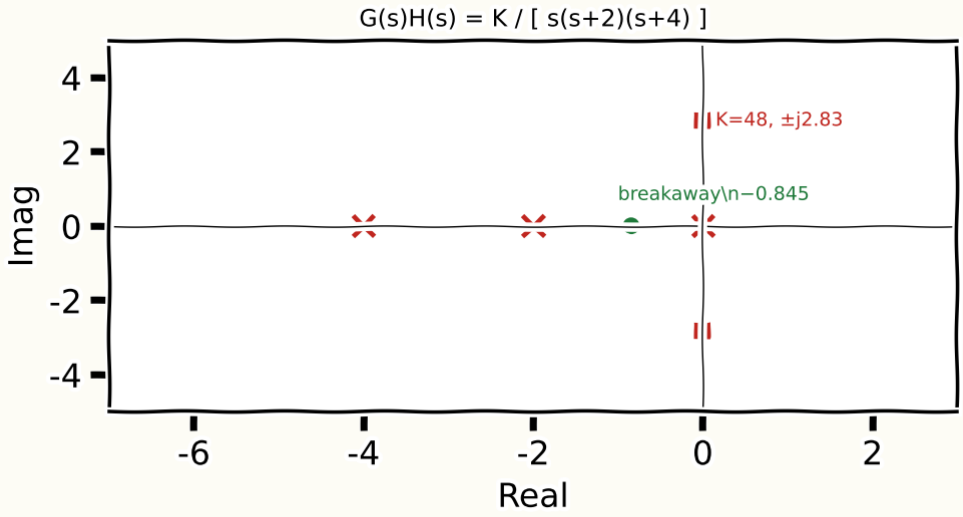
Magnitude condition: $|K\,G(s)H(s)| = 1 \Rightarrow K = \prod|s - \text{poles}| \,/\, \prod|s - \text{zeros}|$

Construction rules (P = no. of open loop poles, Z = no. of zeros)

#	Rule
1	Locus starts ($K=0$) at open loop poles and ends ($K=\infty$) at zeros; $P-Z$ branches go to infinity (zeros at infinity). No. of branches = $\max(P, Z)$.
2	Locus is symmetric about the real axis.
3	A point on the real axis lies on the locus if the number of poles + zeros to its right is **odd**.
4	Asymptotes: angles $\theta = (2q+1)180^\circ/(P-Z)$; centroid $\sigma_A = (\Sigma \text{ poles} - \Sigma \text{ zeros})/(P - Z)$.
5	Breakaway / break-in points: solve $dK/ds = 0$ with $K = -1/[G(s)H(s)]$; accept real roots lying on the locus.
6	Angle of departure from complex pole: $180^\circ - \Sigma$(angles of other poles) + Σ(angles of zeros); angle of arrival similarly.
7	Imaginary axis crossing: from Routh array (K_{max} with row of zeros) → auxiliary equation gives ω.
8	Gain K at any point s_1 on the locus from the magnitude condition.

Example: G(s)H(s) = K / [s(s+2)(s+4)]

- $P = 3, Z = 0 \rightarrow 3$ branches, all go to ∞. Real axis: $[0, -2]$ and $(-4, -\infty)$.
- Asymptotes: $\theta = 60°, 180°, 300°$; centroid $\sigma_A = (0 - 2 - 4)/3 = -2$.
- Breakaway: $K = -(s^3 + 6s^2 + 8s)$; $dK/ds = 0 \Rightarrow 3s^2 + 12s + 8 = 0 \Rightarrow s = -0.845$ (valid), -3.155 (not on locus).
- Crossing: Routh gives $K = 48$, $6s^2 + 48 = 0 \Rightarrow s = \pm j2.83$. System stable for $0 < K < 48$.



Effect of adding poles and zeros to G(s)H(s) on the root locus

Addition	Effect on root locus	Effect on system
Pole (LHP)	Locus bends/shifts to the **right**; asymptote angles change; centroid moves right	Reduces relative stability, increases oscillation; K_{max} decreases
Pole at origin	Increases type; locus moves right	Better steady-state accuracy, poorer stability
Zero (LHP)	Locus bends to the **left** (toward the zero); centroid moves left	Increases stability and damping; faster response
Zero in RHP	Moves locus to right	Reduces stability (non-minimum phase)

Important Questions

- Derive expressions for t_r, t_p, M_p, t_s of an underdamped second order system. (10M)
- Explain the step response of a second-order system for various damping ratios. (5M)
- Define K_p, K_v, K_a and find steady-state errors for a given system. (10M)
- State Routh's criterion; special cases with examples; find K for stability. (10M)
- Sketch the root locus for the given G(s)H(s); find breakaway, $j\omega$-crossing and K_{max}. (10M)
- Effect of adding poles and zeros on root locus; effects of PD and PI controllers. (5M)

1. Introduction to Frequency Response Analysis

Frequency response *is the steady-state response of a system to a sinusoidal input. For a stable LTI system with input* $r(t) = A \sin \omega t$*, the output is* $c_{ss}(t) = A|G(j\omega)| \sin(\omega t + \angle G(j\omega))$*: same frequency, magnitude multiplied by* $|G(j\omega)|$ *and phase shifted by* $\angle G(j\omega)$*.*

$$\boldsymbol{G(j\omega) = G(s)|_{s=j\omega} = M(\omega)\ \angle\varphi(\omega)\ ,\ M = |G(j\omega)|\ ,\ \varphi = \angle G(j\omega)}$$

- ***Advantages:*** *(i) experimental data can be obtained easily; (ii) T.F. can be found from measurements; (iii) stability (relative) is read directly without finding roots; (iv) noise and disturbance effects are visible; (v) design of compensators is simple (graphical); (vi) applicable to systems with transport delay.*
- ***Disadvantages:*** *indirect link with time response; needs a stable open loop for Bode-based conclusions (use Nyquist otherwise).*

2. Frequency Domain Specifications

Specification	Meaning
Resonant peak M_r	Maximum value of $\|M(j\omega)\|$ of closed loop; indicates relative stability (large M_r → large overshoot)
Resonant frequency ω_r	Frequency at which M_r occurs; indicates speed of response
Bandwidth ω_b	Frequency at which $\|M\|$ falls to 0.707 (−3 dB) of its zero-frequency value; large bandwidth → fast response but more noise
Cut-off rate	Slope of log-magnitude beyond ω_b; ability to discriminate noise
Gain crossover ω_{gc}	Frequency where $\|G(j\omega)H(j\omega)\| = 1$ (0 dB)
Phase crossover ω_{pc}	Frequency where $\angle G(j\omega)H(j\omega) = -180°$
Gain margin (GM)	$GM = 1/\|G(j\omega_{pc})H(j\omega_{pc})\|$; in dB: $GM = -20 \log\|GH\|$ at ω_{pc}
Phase margin (PM)	$PM = 180° + \angle G(j\omega_{gc})H(j\omega_{gc})$

Relations for a standard second-order system

$$M_r = 1 / (2\zeta\sqrt{(1-\zeta^2)}) \; (\zeta < 0.707) \; ; \; \omega_r = \omega_n\sqrt{(1-2\zeta^2)}$$

$$\omega_b = \omega_n [\, (1-2\zeta^2) + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}\,]^{1/2}$$

$$PM = \tan^{-1} [\, 2\zeta / \sqrt{(\sqrt{(1+4\zeta^4)} - 2\zeta^2)}\,] \approx 100\zeta \text{ degrees (for } \zeta \leq 0.6)$$

★ Time vs frequency: large M_r ⊠ large M_p; large ω_b ⊠ small t_r; good design: PM = 30°–60°, GM > 6 dB (≥ 2).

3. Bode Diagrams

A Bode plot has two graphs against log ω: (a) magnitude 20 log|G(jω)| in dB and (b) phase angle ∠G(jω) in degrees.

Decade = ratio 10 in frequency; **octave** = ratio 2. 20 dB/decade ≈ 6 dB/octave.

Standard factors

Factor	Magnitude (asymptotic)	Phase
K	$20 \log K$, flat	$0°$ $(K>0)$
$1/(j\omega)^N$	$-20N$ dB/dec, passes 0 dB at $\omega = 1$	$-90°N$
$(j\omega)$	+20 dB/dec through 0 dB at $\omega = 1$	$+90°$
$1/(1 + j\omega T)$	0 dB up to $\omega_c = 1/T$ then -20 dB/dec (actual error -3 dB at ω_c)	$0° \to -45°$ at $\omega_c \to -90°$
$(1 + j\omega T)$	0 dB up to $1/T$, then +20 dB/dec	$0° \to +45° \to +90°$
$1/[1 + 2\zeta(j\omega/\omega_n) + (j\omega/\omega_n)^2]$	0 dB up to ω_n, then -40 dB/dec; resonant peak $-20 \log 2\zeta$ at ω_n for small ζ	$0° \to -90°$ at $\omega_n \to -180°$
$e^{-j\omega T}$ (delay)	0 dB	$-\omega T$ rad (decreasing without bound)

Procedure to draw Bode magnitude plot

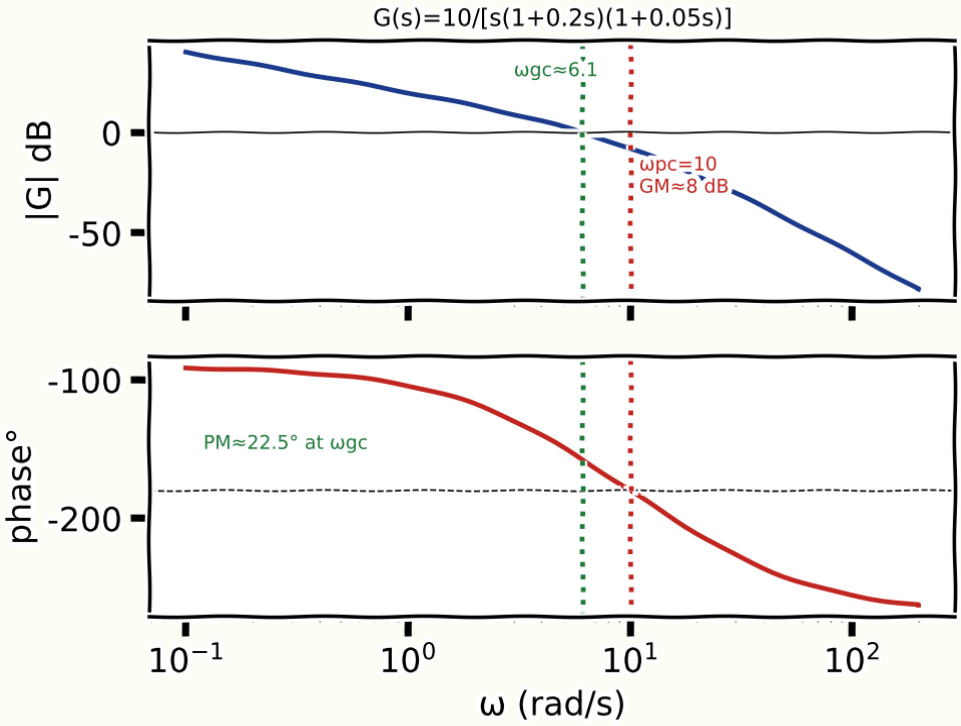
- Write $G(j\omega)$ in standard (time-constant) form; identify K, type N, corner frequencies ($1/T$, ω_n).
- Mark corner frequencies on log scale in increasing order.
- Start: for type 0 draw horizontal line at $20 \log K$; for type N draw line of slope $-20N$ dB/dec passing through the point ($\omega = 1$, $20 \log K$).

- At each corner frequency change the slope: −20 dB/dec for a pole, +20 for a zero (−40 for a quadratic pole pair).
- Phase: evaluate $\varphi(\omega) = \Sigma$ angles at many frequencies (or sketch each factor and add).

Worked example

$G(s) = 10/[\, s(1 + 0.2s)(1 + 0.05s)\,]$. $K = 10$, type 1; corners: $\omega_1 = 1/0.2 = 5$, $\omega_2 = 1/0.05 = 20$ rad/s.

Range of ω	Slope	Remark
$\omega < 5$	−20 dB/dec	At $\omega = 1$ magnitude = 20 log 10 = 20 dB
$5 < \omega < 20$	−40 dB/dec	Gain at 5: 20 − 20 log 5 = 6 dB
$\omega > 20$	−60 dB/dec	Gain at 20 = 6 − 40 log 4 = −18 dB



$$\text{Phase: } \varphi(\omega) = -90^\circ - \tan^{-1}(0.2\omega) - \tan^{-1}(0.05\omega)$$

$$\omega_{pc}: \tan^{-1}(0.2\omega) + \tan^{-1}(0.05\omega) = 90^\circ \Rightarrow 0.01\omega^2 = 1$$

$$\Rightarrow \omega_{pc} = 10 \text{ rad/s}$$

$$|G(j10)| = 10/(10 \cdot \sqrt{5} \cdot \sqrt{1.25}) = 0.4 \Rightarrow GM = 20 \log(1/0.4)$$

$$= 7.96 \text{ dB}$$

$$\omega_{gc} \approx 6.1 \text{ rad/s} \Rightarrow PM = 180^\circ - 90^\circ - 50.6^\circ - 16.9^\circ$$

$$\approx 22.5^\circ$$

*Both GM and PM are positive → **stable** (but PM is low → oscillatory; add lead compensation to improve).*

4. Frequency Domain Specifications and T.F. from the Bode Diagram

Gain margin and phase margin from the Bode plot

- ***PM:*** *read the phase at ω_{gc} (where magnitude curve crosses 0 dB); PM = 180° + $\varphi(\omega_{gc})$.*
- ***GM:*** *read the magnitude at ω_{pc} (phase = −180°); GM = −(magnitude in dB).*
- ***Stability from Bode:*** *GM > 0 dB and PM > 0 ⇒ stable; GM = 0 and PM = 0 ($\omega_{gc} = \omega_{pc}$) ⇒ marginally stable; negative GM/PM ⇒ unstable. For stability, $\omega_{gc} < \omega_{pc}$ (minimum phase systems).*

Determination of transfer function from Bode plot (minimum-phase)

- Initial slope of the magnitude plot gives system type: 0 dB/dec → type 0, −20 → type 1, −40 → type 2.
- Corner frequencies are where slope changes: change of −20 dB/dec → pole $(1 + s/\omega_c)$ in denominator; +20 dB/dec → zero in numerator; −40 → quadratic factor (look at peak to get ζ: $M_r = 1/(2\zeta\sqrt{(1-\zeta^2)})$).
- Find K: type 0 → $20 \log K$ = low-frequency gain; type 1 → the initial −20 dB/dec line (extended) cuts the 0 dB axis at $\omega = K$ (or magnitude at $\omega = 1$ is $20 \log K$); type 2 → cuts at $\omega = \sqrt{K}$.
- Write $G(s) = K \prod(1 + s/z_i) / [\, s^N \prod(1 + s/p_j)\,]$; check the phase curve to confirm minimum phase (and delay if phase falls extra).

Example: initial slope −20 dB/dec, cuts 0 dB at $\omega = 2$ with a further slope change to −40 at $\omega = 10$: $K = 2 \rightarrow G(s) = 2/[s(1 + s/10)] = 20/[s(s + 10)]$.

5. Stability Analysis from Bode Plots

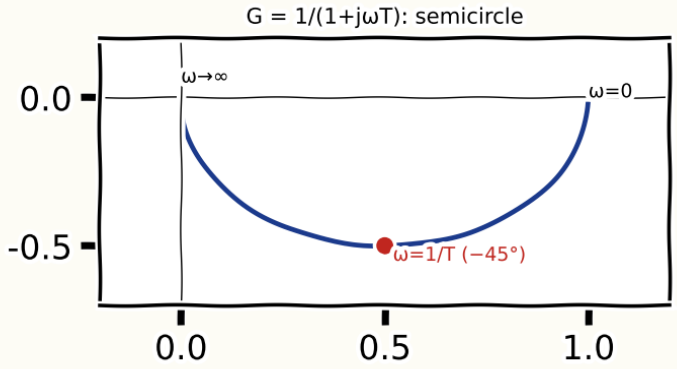
Case	Bode observation	Conclusion
Stable	$\omega_{gc} < \omega_{pc}$, $GM > 0$ dB, $PM > 0$	Closed loop stable
Marginal	$\omega_{gc} = \omega_{pc}$, $GM = 0$, $PM = 0$	Sustained oscillations
Unstable	$\omega_{gc} > \omega_{pc}$, $GM < 0$, $PM < 0$	Growing oscillations

- Increasing K raises the magnitude curve (phase unchanged) → ω_{gc} moves right → PM and GM decrease. Gain for marginal stability: $K_m = K \times 10^{GM/20}$ (e.g., $K = 10$, $GM = 7.96$ dB → $K_{marginal} \approx 25$).
- Phase margin relates to damping: $\zeta \approx PM/100$; GM relates to how much gain increase is tolerable.
- Bode method is not suitable for conditionally stable or open-loop unstable systems → use Nyquist.

★ Merits of Bode plots: easy to draw (straight-line asymptotes), gain & phase on one sheet, multiplication of T.F.s becomes addition, easy compensator design, T.F. from experiments. Demerit: only for minimum-phase systems in the simple form of stability test.

6. Polar Plots

*The **polar plot** of G(jω) is the locus of the tip of the phasor G(jω) (real part vs imaginary part, or magnitude vs phase) as ω varies from 0 to ∞. It gives magnitude and phase on a single plot.*



General shape rules (T.F. with n poles, m zeros, type N)

Type N	Start (ω = 0)	End (ω → ∞)
0	Finite magnitude K on the positive real axis (0°)	Magnitude 0, approaching angle $-90°(n-m)$
1	Magnitude ∞ at angle −90°; asymptote parallel to imaginary axis	Same as above
2	Magnitude ∞ at −180°	$-90°(n-m)$
3	Magnitude ∞ at −270°	$-90°(n-m)$

Procedure

- Write $G(j\omega) = M(\omega)\angle\varphi(\omega)$; find M and φ at $\omega = 0$ and $\omega = \infty$.
- Find the intersection with the real axis (set $Im = 0 \rightarrow \omega_{pc}$, real value $\rightarrow$ GM) and with the imaginary axis ($Re = 0$).
- Plot some intermediate points; join smoothly.

Example: $G(s) = 1/(1 + sT)$: $G(j\omega) = 1/\sqrt{(1+\omega^2T^2)}\ \angle -\tan^{-1}\omega T$; $\omega = 0 \rightarrow 1\angle 0°$; $\omega = 1/T \rightarrow 0.707\angle -45°$; $\omega = \infty \rightarrow 0\angle -90°$. Locus is a **semicircle** in the 4th quadrant of diameter 1.

7. Nyquist Plots and Stability Analysis

The **Nyquist plot** is the polar plot for ω from $-\infty$ to $+\infty$ (the part for negative ω is the mirror image about the real axis). Stability of the closed loop is decided from how it encircles the critical point $\mathbf{-1 + j0}$.

Nyquist contour and mapping

- Nyquist contour: a semicircle of infinite radius enclosing the entire right half s-plane (clockwise direction) with the imaginary axis as its diameter.
- If $G(s)H(s)$ has poles on the $j\omega$ axis (e.g., at origin) indent the contour around them with a small semicircle of radius $\varepsilon \rightarrow 0$ in the RHP. For N poles at origin, the infinite-radius arc in the $G(s)H(s)$ plane covers $N \times 180°$ **clockwise**, from the $\omega = 0^-$ end to $\omega = 0^+$ end.

- *The contour of the 1 + GH function encircling the origin ⇔ the GH plot encircling −1.*

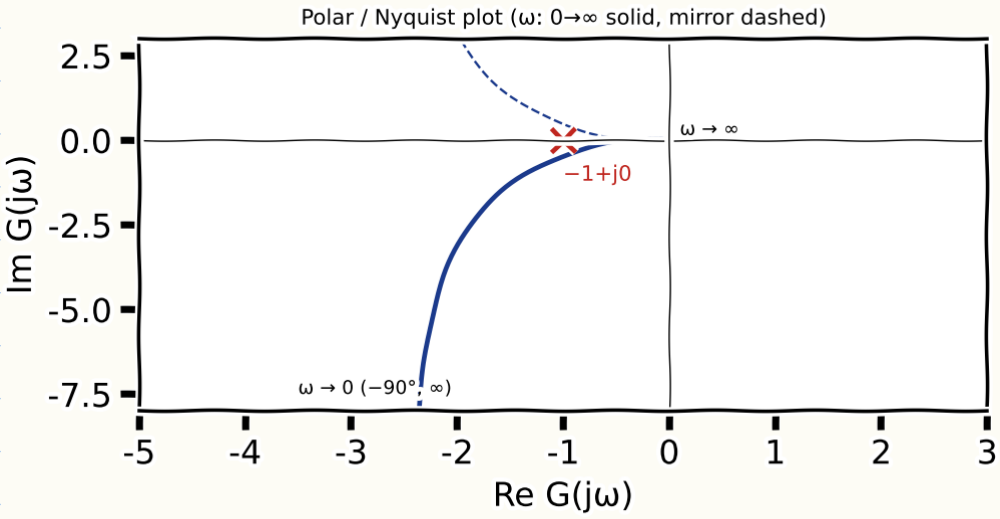
Nyquist stability criterion

$$Z = P - N$$

P = number of open-loop poles in the RHP ; N = number of counter-clockwise encirclements of −1 + j0 by the Nyquist plot ; Z = number of closed-loop poles in the RHP

The closed loop is stable iff Z = 0, i.e., N = P

- *If the open loop is stable (P = 0): closed loop is stable iff the Nyquist plot* ***does not encircle*** *−1 + j0.*
- *If the plot passes through −1 + j0 → marginally stable (poles on jω axis). Clockwise encirclements ⇒ N negative ⇒ Z > 0 unstable.*



For the example G(s) = 10/[s(1 + 0.2s)(1 + 0.05s)], P = 0; the plot crosses the negative real axis at −0.4 (at ω = 10) which is to

the right of -1 → no encirclement → $N = 0 = P$ → ***stable****. Gain margin = 1/0.4 = 2.5 (7.96 dB); a gain increase beyond 2.5 pushes the crossing past -1 → unstable.*

Gain and phase margin from the Nyquist plot

GM = $1 / |G(j\omega_{pc})H(j\omega_{pc})|$ (point where the plot cuts the negative real axis)

PM = $180° + \angle G(j\omega_{gc})H(j\omega_{gc})$ (angle at the point where plot cuts the unit circle)

Feature	Bode plot	Polar plot	Nyquist plot
Axes	Two (dB, degrees vs log ω)	One (Re vs Im)	One (Re vs Im, $-\infty \to \infty$)
Stability	GM, PM from curves	GM, PM	Encirclement of -1 ($Z = P - N$)
Open-loop unstable systems	Not suitable	Not suitable	Suitable
Ease of drawing	Easiest	Moderate	Needs care at origin/infinite arcs

Important Questions

- Define M_r, ω_r, ω_b, GM, PM; derive M_r and ω_r for a 2nd-order system. (10M)

- *Draw the Bode plot for the given G(s)H(s); find GM, PM and comment on stability. (10M)*
- *Determine the T.F. from the given Bode magnitude plot. (5M/10M)*
- *Sketch the polar plot for the given T.F. (5M)*
- *State Nyquist criterion; draw the Nyquist plot and find the range of K for stability. (10M)*

1. Concepts of State, State Variables and State Model

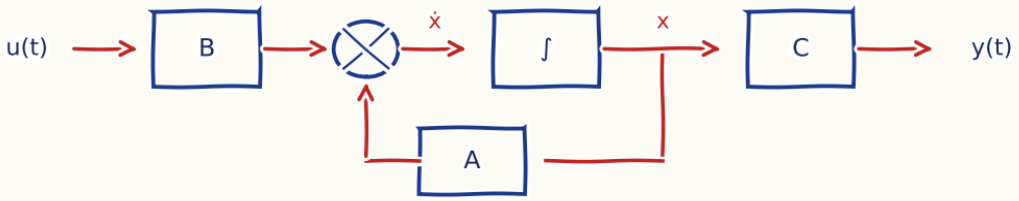
Classical (transfer function) methods deal with single-input single-output LTI systems with zero initial conditions. ***State space analysis*** *(modern control) is a time-domain method that handles MIMO, time-varying, non-linear systems and non-zero initial conditions, and is convenient for computers.*

Term	Definition
State	Minimum set of variables whose values at $t = t_0$, together with inputs for $t \geq t_0$, determine the behaviour of the system for all $t \geq t_0$
State variables	The variables $x_1(t), \ldots, x_n(t)$ that make up the state; n = order of the system (number of energy-storing elements / integrators)
State vector	Column vector $x(t) = [x_1 \ldots x_n]^T$
State space	n-dimensional space whose axes are the state variables; the system trajectory lies in it
State equation	First-order differential equations for the derivatives of the states
Output equation	Algebraic equation giving outputs in terms of states and inputs

State equation: $x'(t) = A\,x(t) + B\,u(t)$

Output equation: $y(t) = C\,x(t) + D\,u(t)$

- $x : n \times 1$ state vector; $u : r \times 1$ input vector; $y : m \times 1$ output vector.
- $A : n \times n$ system (state) matrix; $B : n \times r$ input matrix; $C : m \times n$ output matrix; $D : m \times r$ direct-transmission (feed-forward) matrix (usually zero).



Advantages of state space over transfer function

- Applies to MIMO, non-linear and time-varying systems; initial conditions included.
- Gives internal behaviour (all states), not only input–output; suitable for optimal control, digital computers.
- Can handle controllability and observability analysis; design by pole placement.

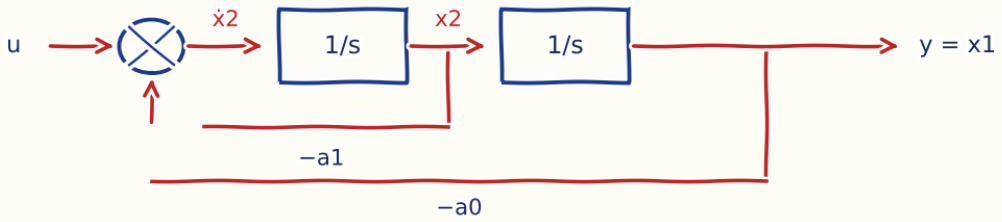
2. Derivation of State Models

(a) From a differential equation / transfer function – phase variables

For $d^n y/dt^n + a_{n-1} d^{n-1} y/dt^{n-1} + \ldots + a_0 y = u$ choose $x_1 = y$, $x_2 = y'$, …, $x_n = y^{(n-1)}$. Then $x'_1 = x_2$, …, $x'_n = -a_0 x_1 - \ldots - a_{n-1} x_n + u$.

$A = [\,0\,1\,0\,;\,0\,0\,1\,;\,-a_0\,-a_1\,-a_2\,]$ (3rd order, rows separated by ;)

$B = [\,0\,;\,0\,;\,1\,]$, $C = [\,1\,0\,0\,]$, $D = 0$ (companion / controllable canonical form)



If the T.F. has numerator, $G(s) = (b_2 s^2 + b_1 s + b_0)/(s^3 + a_2 s^2 + a_1 s + a_0)$, A and B stay the same and **$C = [\,b_0\,b_1\,b_2\,]$**.

Example

$y''' + 6y'' + 11y' + 6y = u$:

$A = [\,0\,1\,0\,;\,0\,0\,1\,;\,-6\,-11\,-6\,]$, $B = [\,0\,;\,0\,;\,1\,]$, $C = [\,1\,0\,0\,]$

(b) From an electrical network

Choose capacitor voltages and inductor currents as state variables. Series R–L–C, input v_i, output v_c; $x_1 = v_c$, $x_2 = i_L$:

$$x'_1 = x_2/C \; ; \; x'_2 = -x_1/L - (R/L)x_2 + v_i/L \; ; \; y = x_1$$
$$A = [\, 0 \;\; 1/C \; ; \; -1/L \;\; -R/L \,] \; , \; B = [\, 0 \; ; \; 1/L \,] \; , \; C = [\, 1 \;\; 0 \,]$$

(c) From a block diagram

- Redraw the diagram so that each block is a simple first-order block $1/(s + a)$ or an integrator $1/s$.
- Assign a state variable to the **output of each integrator / first-order block**.
- Write x' for each block (e.g., block $1/(s+a)$: $x' = -a\,x + \text{input}$) and express the input of each block using states and the reference input.
- Write the output y as a combination of states.

Example: unity-feedback system, $G(s) = 10/[s(s+2)]$ split as $e \rightarrow [10/(s+2)] \rightarrow x_2 \rightarrow [1/s] \rightarrow x_1 = c$, with $e = r - x_1$:

$$x'_2 = -2x_2 + 10(r - x_1) \; , \; x'_1 = x_2$$
$$A = [\, 0 \;\; 1 \; ; \; -10 \;\; -2 \,] \; , \; B = [\, 0 \; ; \; 10 \,] \; , \; C = [\, 1 \;\; 0 \,] \; (\text{check: } |sI - A| = s^2 + 2s + 10 \; \square)$$

(d) Parallel (diagonal) and Jordan forms

Expand $G(s)$ by partial fractions $G(s) = \Sigma\, c_i/(s - p_i)$ for distinct poles. Each term gives $x'_i = p_i x_i + u$; $y = \Sigma\, c_i x_i \rightarrow A = \text{diag}(p_1, \ldots, p_n)$ (decoupled states). For repeated poles the Jordan canonical form has 1's on the superdiagonal.

Transfer function from state model

$$\mathbf{G(s) = Y(s)/U(s) = C\,(sI - A)^{-1}\,B + D}$$

Characteristic equation: $|sI - A| = 0$; eigenvalues of A = poles of the system

Check: $A = [\,0\ 1\,;\,-2\ -3\,]$, $B = [\,0\,;\,1\,]$, $C = [\,1\ 0\,]$: $(sI - A)^{-1} = 1/[(s+1)(s+2)]\,[\,s+3\ 1\,;\,-2\ s\,] \Rightarrow G(s) = 1/[(s+1)(s+2)]$.

3. Diagonalization

Diagonalising A decouples the state equations, making analysis (and e^{At}) easy. Use the transformation $x = M\,z$ where M is the **modal matrix** (columns = eigenvectors):

$$\mathbf{z' = \Lambda\, z + M^{-1}B\,u\,,\ y = C\,M\,z + D\,u\,,\ \Lambda = M^{-1}\,A\,M = diag(\lambda_1, \ldots, \lambda_n)}$$

Steps

- Find eigenvalues λ_i from $|\lambda I - A| = 0$.
- For each λ_i solve $(\lambda_i I - A)\,m_i = 0$ for the eigenvector m_i.
- Form $M = [\,m_1\ m_2 \ldots m_n\,]$ and compute $\Lambda = M^{-1}AM$.

- **Shortcut:** *for a companion (phase-variable) matrix with distinct eigenvalues, M is the Vandermonde matrix: $M = [\,1\ 1\ \dots\ 1\,;\,\lambda_1\ \lambda_2\ \dots\ \lambda_n\,;\,\lambda_1^2\ \lambda_2^2\ \dots\ \lambda_n^2\,;\,\dots\,]$.*

Example

$A = [\,0\ 1\,;\,-2\ -3\,]$: $|\lambda I - A| = \lambda^2 + 3\lambda + 2 = 0 \Rightarrow \lambda_1 = -1, \lambda_2 = -2$.

$$m_1 = [\,1\,;\,-1\,],\ m_2 = [\,1\,;\,-2\,] \Rightarrow M = [\,1\ 1\,;\,-1\ -2\,],$$
$$M^{-1} = [\,2\ 1\,;\,-1\ -1\,]$$
$$\Lambda = M^{-1}AM = [\,-1\ 0\,;\,0\ -2\,]$$

★ *Repeated eigenvalues → generalised eigenvectors → Jordan form $J = M^{-1}AM$. If eigenvalues are complex conjugate use real modal form with 2×2 blocks $[\,\sigma\ \omega\,;\,-\omega\ \sigma\,]$.*

4. Solution of Time-Invariant State Equations

Homogeneous equation ($u = 0$): $x' = Ax$, $x(0) = x_0$

$$x(t) = e^{At}\,x(0) = \Phi(t)\,x(0)$$
$$e^{At} = I + At + A^2t^2/2! + A^3t^3/3! + \dots$$
$$\Phi(t) = L^{-1}\{\,(sI - A)^{-1}\,\}$$

Complete solution (with input)

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau$$

$$y(t) = Cx(t) + Du(t)$$

$$\text{Laplace form: } X(s) = (sI - A)^{-1}x(0) + (sI - A)^{-1}BU(s)$$

The first term is the ***zero-input (free) response*** *and the second is the* ***zero-state (forced) response*** *= convolution of $\Phi(t)B$ with $u(t)$.*

Methods of computing e^{At}

- ***Laplace transform:*** $\Phi(t) = L^{-1}[(sI - A)^{-1}]$.
- ***Diagonalisation:*** $e^{At} = M\,e^{\Lambda t}\,M^{-1}$ *with* $e^{\Lambda t} = \mathrm{diag}(e^{\lambda_1 t}, \ldots, e^{\lambda_n t})$.
- ***Cayley–Hamilton:*** $e^{At} = \alpha_0 I + \alpha_1 A + \ldots + \alpha_{n-1}A^{n-1}$, *where* $e^{\lambda_i t} = \alpha_0 + \alpha_1\lambda_i + \ldots$ *(one equation per eigenvalue).*
- ***Power series*** *(truncated; useful when A is nilpotent).*

Worked example ($A = [\,0\;1\,;\,-2\;-3\,]$)

$(sI - A)^{-1} = 1/[(s+1)(s+2)]\,[\,s+3\;1\,;\,-2\;s\,]$. *By partial fractions:*

$$\Phi(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

For $B = [\,0\,;\,1\,]$, unit-step input, $x(0) = 0$: $x_1(t) = \int (e^{-(t-\tau)} - e^{-2(t-\tau)})d\tau$ *and* $x_2 = x'_1$:

$$x_1(t) = \tfrac{1}{2} - e^{-t} + \tfrac{1}{2}e^{-2t},\quad x_2(t) = e^{-t} - e^{-2t}$$

5. State Transition Matrix and Its Properties

The **state transition matrix (STM)** $\Phi(t) = e^{At}$ describes how the state propagates from the initial state: $x(t) = \Phi(t - t_0)x(t_0)$ for the zero-input case. It satisfies $\Phi'(t) = A\Phi(t)$ with $\Phi(0) = I$.

#	Property	Meaning
1	$\Phi(0) = I$	Identity at $t = 0$
2	$\Phi^{-1}(t) = \Phi(-t)$	Inverse = time reversal
3	$\Phi(t_1 + t_2) = \Phi(t_1)\Phi(t_2) = \Phi(t_2)\Phi(t_1)$	Semigroup / commutative
4	$[\Phi(t)]^k = \Phi(kt)$	Power property
5	$\Phi(t_2 - t_1)\Phi(t_1 - t_0) = \Phi(t_2 - t_0)$	Transition property
6	$d\Phi/dt = A\Phi(t) = \Phi(t)A$	Derivative; $\Phi'(0) = A$
7	$\Phi(t) = M\, e^{\Lambda t}\, M^{-1}$	Diagonalisation

Proof of (2): $\Phi(t)\Phi(-t) = e^{At}e^{-At} = e^{0} = I \Rightarrow \Phi^{-1}(t) = \Phi(-t)$.

For the example above, check $\Phi(0) = [\ 2-1\ 1-1\ ;\ -2+2\ -1+2\ ] = [\ 1\ 0\ ;\ 0\ 1\ ]$ ☒ and $\Phi'(0) = [\ -2+2\ -1+2\ ;\ 2-4\ 1-4\ ] = [\ 0\ 1\ ;\ -2\ -3\ ] = A$ ☒.

6. Controllability and Observability

Controllability

A system is ***completely state controllable*** *if there exists a control u(t) that can transfer any initial state $x(t_0)$ to any final state in finite time.*

> **Kalman's test: $Q_c = [\, B \; AB \; A^2B \; \dots \; A^{n-1}B \,]$;**
> **controllable iff $\text{rank}(Q_c) = n$ ($|Q_c| \neq 0$ for single input)**

Gilbert's test *(diagonal form, distinct eigenvalues): controllable iff the matrix $M^{-1}B$ has* ***no zero row*** *(a zero row means that mode cannot be affected by the input).*

Observability

A system is ***completely observable*** *if every initial state $x(t_0)$ can be determined from the knowledge of the output y(t) and input over a finite interval.*

> **Kalman's test: $Q_o = [\, C^T \; A^TC^T \; (A^T)^2C^T \; \dots \; (A^T)^{n-1}C^T \,]$;**
> **observable iff $\text{rank}(Q_o) = n$**

Gilbert's test: *in diagonal form, observable iff the matrix CM has* ***no zero column****.*

Examples

(i) $A = [\,0\ 1\,;\,-2\ -3\,]$, $B = [\,0\,;\,1\,]$, $C = [\,1\ 0\,]$: $AB = [\,1\,;\,-3\,] \Rightarrow Q_c = [\,0\ 1\,;\,1\ -3\,]$, $|Q_c| = -1 \neq 0 \rightarrow$ **controllable**. $A^T C^T = [\,0\,;\,1\,] \Rightarrow Q_o = [\,1\ 0\,;\,0\ 1\,]$, rank 2 → **observable**.

(ii) $A = [\,-1\ 0\,;\,0\ -2\,]$, $B = [\,1\,;\,0\,]$: $AB = [\,-1\,;\,0\,]$, $Q_c = [\,1\ -1\,;\,0\ 0\,]$, $|Q_c| = 0 \rightarrow$ **not controllable** (second mode x_2 unaffected by u).

(iii) $A = [\,1\ 0\,;\,0\ -1\,]$, $B = [\,1\,;\,1\,]$, $C = [\,0\ 1\,]$: $Q_c = [\,1\ 1\,;\,1\ -1\,] \rightarrow$ rank 2 (controllable); $Q_o = [\,C^T\ A^T C^T\,] = [\,0\ 0\,;\,1\ -1\,] \rightarrow$ rank 1 → **not observable**.

Point	Controllability	Observability
Question	Can input drive all states?	Can output reveal all states?
Test matrix	$[\,B\ AB\ \ldots\ A^{n-1}B\,]$	$[\,C^T\ A^T C^T\ \ldots\,]$
Gilbert (diagonal)	No zero row in $M^{-1}B$	No zero column in CM
Duality	(A, B) controllable ⇔ (A^T, B^T) observable	(A, C) observable ⇔ (A^T, C^T) controllable

★ **Relation with transfer function: a pole–zero cancellation in G(s) means the system is NOT completely controllable and/or observable. A T.F. gives only the controllable and observable part of the system. Pole placement by state feedback $u = -Kx$ needs controllability.**

7. More Solved Problems (exam type)

Problem 1 – State model from a transfer function with zeros

$G(s) = (s + 3)/(s^3 + 6s^2 + 11s + 6)$. Phase-variable form: $a_0 = 6$, $a_1 = 11$, $a_2 = 6$; numerator $b_0 = 3$, $b_1 = 1$, $b_2 = 0$.

$$\mathbf{A = [\,0\,1\,0\,;\,0\,0\,1\,;\,-6\,-11\,-6\,],\ B = [\,0\,;\,0\,;\,1\,],\ C = [\,3\,1\,0\,],\ D = 0}$$

Poles: $s^3 + 6s^2 + 11s + 6 = (s+1)(s+2)(s+3) \rightarrow -1, -2, -3$ (all in LHP → stable). The numerator zero at -3 cancels the pole at -3, so the system is controllable but **not observable** in this realisation (or vice versa in another).

Problem 2 – Diagonalisation of a 3rd-order system

For the A above, eigenvalues are $-1, -2, -3$ (distinct) and A is in companion form, so the modal matrix is the Vandermonde matrix:

$$\mathbf{M = [\,1\,1\,1\,;\,-1\,-2\,-3\,;\,1\,4\,9\,] \Rightarrow \Lambda = M^{-1}AM = diag(-1, -2, -3)}$$

In the new coordinates $z = M^{-1}x$ each mode evolves independently: $z_i(t) = e^{\lambda_i t} z_i(0) + \int e^{\lambda_i (t-\tau)} (M^{-1}B)_i \, u \, d\tau$.

Problem 3 – e^{At} by Cayley–Hamilton

$A = [\,0\ 1\,;\,-2\ -3\,]$, eigenvalues $-1, -2$. For $n = 2$: $e^{At} = \alpha_0 I + \alpha_1 A$ with

$$e^{-t} = \alpha_0 - \alpha_1,\ e^{-2t} = \alpha_0 - 2\alpha_1$$
$$\Rightarrow \alpha_1 = e^{-t} - e^{-2t},\ \alpha_0 = 2e^{-t} - e^{-2t}$$

Hence $e^{At} = [\,\alpha_0\ \alpha_1\,;\,-2\alpha_1\ \alpha_0 - 3\alpha_1\,] = [\,2e^{-t} - e^{-2t}\ e^{-t} - e^{-2t}\,;\,-2e^{-t} + 2e^{-2t}\ -e^{-t} + 2e^{-2t}\,]$ — the same as the Laplace method ☒.

Problem 4 – Zero-input response

With $x(0) = [\,1\,;\,0\,]$ and the same A: $x(t) = \Phi(t)x(0) = [\,2e^{-t} - e^{-2t}\,;\,-2e^{-t} + 2e^{-2t}\,]$; both states decay to zero because the eigenvalues are negative.

Stability and the state model

- Asymptotically stable iff all eigenvalues of A (roots of $|sI - A| = 0$) have negative real parts.
- Eigenvalues are invariant under similarity transformation $x = Mz$; the transfer function is also invariant.
- State feedback $u = -Kx$ changes A to $(A - BK)$; if (A, B) is controllable the eigenvalues can be placed anywhere (pole placement) by suitable K.

Quick-revision formulas

$$x' = Ax + Bu\ ,\ y = Cx + Du \mid G(s) = C(sI - A)^{-1}B + D$$

$$\Phi(t) = e^{At} = L^{-1}[(sI - A)^{-1}] = M\, e^{\Lambda t}\, M^{-1}$$

$$x(t) = \Phi(t)x(0) + \int \Phi(t - \tau)Bu(\tau)d\tau$$

$$Q_c = [B\ AB \dots A^{n-1}B]\ ,\ Q_o = [C^T\ A^TC^T \dots] \rightarrow rank\ n$$

Important Questions

- *Define state, state variable, state model; obtain state model of a given electrical network / T.F. (5M/10M)*
- *Derive the state model from a block diagram. (5M)*
- *Diagonalise the given matrix A and find the modal matrix. (10M)*
- *Find the state transition matrix and solution x(t) for a given system by Laplace/Cayley–Hamilton method. (10M)*
- *List properties of STM with proof. (5M)*
- *State and test Kalman's controllability and observability criteria for the given system. (10M)*